

9.1 Maximal Margin Classifier

Support Vector Machines

In this chapter, we approach the two-class classification problem in a direct way: we try to find a plane that separates the classes in the feature space.

If we cannot, we make some adjustments:

- ▶ Soften what we mean by “separates”.
- ▶ Enrich and enlarge the feature space to make separation possible.

Hyperplanes

- ▶ In a p -dimensional space, a *hyperplane* is a flat, affine subspace of dimension $p - 1$.
 - ▶ *Affine* indicates the subspace does not need to pass through the origin.
- ▶ In two dimensions, this is a one-dimensional subspace (a line).
- ▶ In three dimensions, a plane.

Hyperplanes

In two dimensions, a hyperplane is defined by

$$\beta_0 + \beta_1 X_1 + \beta_2 X_2 = 0$$

► Any $X = (X_1, X_2)^T$ for which this holds is a point on the hyperplane.

In p dimensions,

$$\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p = 0$$

defines a p -dimensional hyperplane.

Hyperplanes

Now, suppose that X does not satisfy this equation. That is,

$$\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p > 0$$

► then X is to one side of the hyperplane

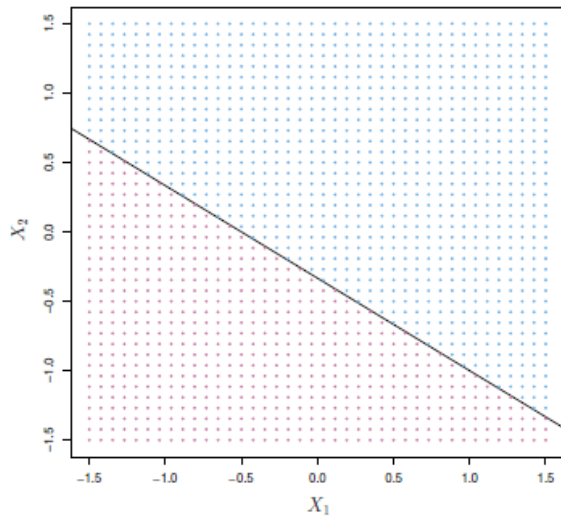
or if

$$\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p < 0$$

► then X is to the other side of the hyperplane.

So we can think of hyperplanes as dividing a p -dimensional space into two parts.

Hyperplanes



Classification Using Hyperplanes

- ▶ Suppose we have some $n \times p$ data matrix X where observations fall into one of two classes.
- ▶ We also have a test observation x^*
- ▶ Our goal is to develop a classifier based on the training data that will correctly classify the test observation using its feature measurements.
- ▶ We have several such approaches already: logistic regression, LDA, classification trees, etc.
- ▶ We now consider an approach using a *separating hyperplane*.

Classification Using Hyperplanes

- ▶ Now suppose it's possible to construct a hyperplane that fully separates the training observations by their class levels.
- ▶ If we label the classes as -1 and 1 , we can say that

$$\beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \cdots + \beta_p X_{ip} > 0$$

if $y_i = 1$ and

$$\beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \cdots + \beta_p X_{ip} < 0$$

if $y_i = -1$ or that a separating hyperplane is such that

$$y_i(\beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \cdots + \beta_p X_{ip}) > 0$$

for all $i = 1, \dots, n$.

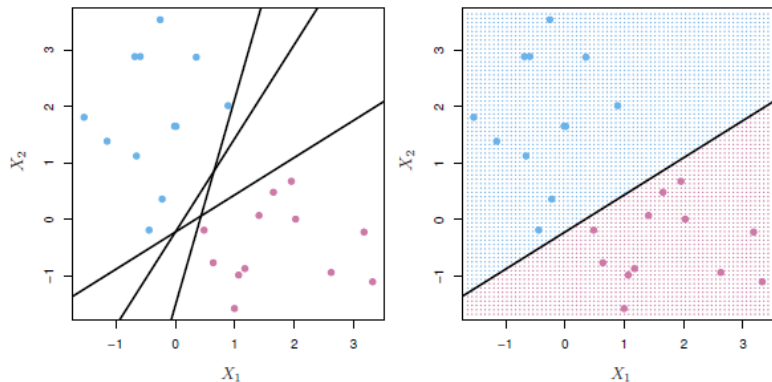
Classification Using Hyperplanes

- ▶ If a separating hyperplane exists, there is a natural classifier:
 - ▶ a test observation is assigned a class depending on which side of the hyperplane it is located.
 - ▶ Note: separating hyperplanes will lead to linear decision boundaries.
- ▶ The *magnitude* of $f(x^*) = \beta_0 + \beta_1 x_1^* + \beta_2 x_2^* + \cdots + \beta_p x_p^*$ acts as a measure of how confident we are in the class assignment
 - ▶ If $f(x^*)$ is far from 0, the point is far from the hyperplane.

The Maximal Margin Classifier

- ▶ If our data can be perfectly separated using a hyperplane, there there are an infinite number of such hyperplanes.
 - ▶ We would like to choose the best one.

Example



Three possible separating hyperplanes are shown on the left.

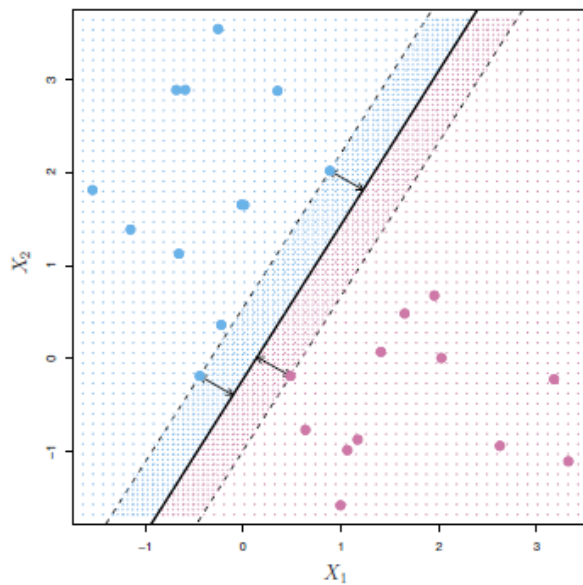
The Maximal Margin Classifier

- ▶ A natural choice of “best” hyperplane is the *maximal margin hyperplane* (or the *optimal separating hyperplane*)
 - ▶ This is the one that is farthest from the training observations.
- ▶ We find the (perpendicular) distance from each training observation to a given separating hyperplane
 - ▶ The smallest such distance is known as the *margin*
- ▶ The *maximal margin classifier* utilizes the hyperplane for which the margin is largest.
 - ▶ That is, the hyperplane has the farthest minimum distance to the training observations.

The Maximal Margin Classifier

- ▶ We hope that a classifier with a large margin on the training data will also have a large margin on the test data.
 - ▶ This should tend to classify correctly.
 - ▶ However, these can lead to overfitting when p is large.

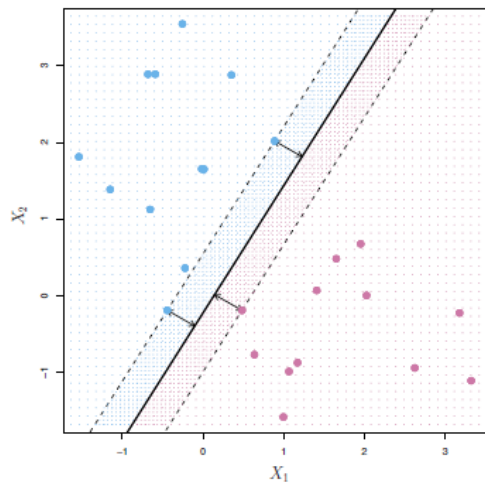
Example



Example

- ▶ The three points which are equidistant from the hyperplane (on the dashed lines) are known as *support vectors*
 - ▶ If any were to be moved, the hyperplane would shift.
 - ▶ The maximal margin hyperplane depends directly on these points (and not the other points).

Construction of the Maximal Margin Classifier



Constrained optimization problem

$$\begin{aligned} &\text{maximize } M \\ &\beta_0, \beta_1, \dots, \beta_p \end{aligned}$$

$$\text{subject to } \sum_{j=1}^p \beta_j^2 = 1,$$

$$y_i(\beta_0 + \beta_1 x_{i1} + \dots + \beta_p x_{ip}) \geq M$$

for all $i = 1, \dots, N$.

Construction of the Maximal Margin Classifier

This turns out to be simpler than it looks!

- ▶ The constraint that

$$y_i(\beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \cdots + \beta_p X_{ip}) \geq M; \quad M > 0$$

guarantees complete and correct separation of the training data.

- ▶ The constraint

$$\sum_{j=1}^p \beta_j^2 = 1$$

forces the perpendicular distance from the i th observation to the hyperplane to be

$$y_i(\beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \cdots + \beta_p X_{ip})$$

- ▶ Then M is the margin for the hyperplane, and the optimization problem maximizes this value.
 - ▶ (The actual optimization problem is outside the scope of this course.)

Maximal Margin Classifiers in R

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library(e1071)  
?svm
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- ▶ We will see this in detail in the next section.